

**Lecture Summary for
Computer Vision A
BY
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The fundamentals of digital image processing were introduced, including how images are represented in different color spaces such as RGB, HSB, and Lab, and how these individual channels can be leveraged for robust features extraction. Various image-processing methods including brightness and contrast normalization, histogram-based thresholding, and binarization through Otsu's method were also proposed.

Filtering techniques ranging from box and Gaussian filters to bilateral filtering, the latter of which preserves edges by combining spatial and intensity information were also discussed. Noise removal methods such as median filtering were demonstrated to effectively eliminate salt-and-pepper noise without excessively blurring important image structures.

Edge detection was explained using first-order derivative operators (such as the Sobel filter for gradient computation) and second-order derivatives (including the Laplacian and Laplacian-of-Gaussian (LoG)), showing how gradient magnitudes and zero-crossing highlight object boundaries in images.

Local feature detecting, focusing on how key points such as corners and interest points are extracted to characterize distinctive structures in images was introduced. Classical detectors such as the Moravec detector - based on intensity variation measured via Sum of Squared Differences (SSD), and the Harris Corner Detector, based on the second-moment matrix (structure tensor) and its eigen values were also covered in the lecture. The Hessian detector was also presented as a curvature-based alternative, using the determinant of the Hessian matrix to identify points of high local curvature.

FAST significantly accelerates corner detection using an optimized segment test and machine-learned decision trees. Feature descriptors such as SIFT, which provide strong scale and rotation invariance via Difference-of-Gaussian (DoG) key point detection and 128-dimensional orientation histograms, and SURF, which achieves similar invariance with higher computational efficiency through integral images and Haar-wavelet responses, were also described in the lecture.

Feature matching forms the foundation for applications such as image stitching, object recognition, and 3D reconstruction. Template matching approaches such as SSD, SAD, NCC and zero-mean NCC were also introduced along with their disadvantages.

Frequency-domain correlation methods using the 2D Fast Fourier Transform (FFT) and Phase-Only Correlation (POC) were presented, offering high precision alignment by comparing only the phase of the Fourier spectra. A significant extension involved converting images to log-polar coordinates, transforming rotation and scale variations into simple translations. This enables POC to accurately estimate rotation, scaling, and translation, making it effective for applications requiring precise registration under geometric transformations.

Panoramic image stitching was also discussed, in which corresponding feature points across images are used to compute the homography that relates one viewpoint to another. Camera intrinsic parameters (focal length, principal point) and extrinsic parameters (rotation and

translation) were explained as the basis for mapping 3D world coordinates into 2D image coordinates.

Building on camera geometry, stereo vision was introduced as a method for estimating depth by comparing images taken from slightly different viewpoints. Epipolar geometry was explained through epipolar planes, epipolar lines, and the fundamental constraint that corresponding points must lie on their respective epipolar lines. Stereo matching algorithms such as Block Matching (BM) and Semi-Global Block Matching (SGBM) were also discussed in the lecture. The real-world applications include small object detection, road-surface recognition, and high-resolution 3D reconstruction for autonomous vehicles were also mentioned in the lecture.

The machine-learning section of the lecture starts with supervised learning, introducing the perceptron as the simplest linear classifier capable of solving linearly separable tasks such as logical AND/OR operations. Multi-Layer perceptrons (MLPs) extend this to non-linear classification using activation functions such as sigmoid, tanh, and ReLU.

Backpropagation and gradient descent were discussed as the core learning mechanisms, with detailed derivations of gradient calculations using computational graphs. Several loss functions including mean squared error, mean absolute error, categorical cross-entropy, and binary cross-entropy were derived step-by-step, emphasizing how gradients are propagated to update network parameters.

Classical machine-learning models such as Support Vector Machines (SVMs), which optimize margin-based decision boundaries, and ensemble-learning methods such as AdaBoost and Random Forests, which combine multiple weak classifiers to produce robust predictions, were also introduced.

The Deep Learning Convolutional Neural Networks (CNNs) are designed to learn spatially localized features from images. Core components such as convolution layers, kernels, feature maps, pooling operations, and fully connected layers were discussed through foundational architectures including LeNet, AlexNet, GoogLeNet, and ResNet. Visualization techniques demonstrated that early layers learn low-level features like edges, while deeper layers capture increasingly complex patterns and semantic representations.

A complete MNIST classification workflow was used to demonstrate practical training steps, including input formatting, Softmax probability computation, cross-entropy loss, and backpropagation through convolutional layers. CNNs were contextualized within modern applications such as image classification, object detection, regression, and segmentation, showing how deep networks have become the standard for computer-vision tasks.

The advanced 3D reconstruction approaches such as Structure-from-Motion (SfM) and Multi-View Stereo (MVS) were also mentioned in the lecture. SfM was introduced as a pipeline for estimating camera poses and generating sparse 3D point clouds from feature correspondences across multiple images and MVS extends this with dense depth-map estimation to produce detailed 3D models, forming the basis of classical multi-view reconstruction system.

Neural Radiance Field (NeRF), which is a modern neural rendering technique that represents scenes as continuous volumetric functions, were also discussed. Using camera poses

from SfM, NeRF learns a mapping from 3D positions and view directions to color and density values. Through volume rendering and ray marching, NeRF enables photorealistic novel-view synthesis. Current active and rapidly evolving directions such as fast NeRF, dynamic scene NeRF, and low computation NeRF were also covered in the lecture.

**Panorama Stitching Using SIFT and
ORB Feature Matching with
Unsupervised Machine Learning
Comparison**

(Project Report for Computer Vision A)

BY

THU TA ZAW

1.INTRODUCTION

Overlapping multiple images to combine into a single seamless one by detecting salient visual features, matching them between different images, estimating geometric transformations, and finally warping multiple images into a common coordinate frame is called image stitching. This can be used in various computer vision applications such as smartphones photography, autonomous driving, and robotics. In this project, image stitching method is used to create complete panorama pictures based on both SIFT (Scale-Invariant Feature Transform) and ORB (Oriented FAST and Rotated BRIEF) feature detectors and descriptors.

2.OBJECTIVES

The primary objectives of this project are:

- To implement the complete panoramas picture using the SIFT and ORB feature detectors and descriptors.
- To compare the performance of SIFT and ORB by using the unsupervised machine learning (K-means clustering).
- To analyze the robustness, accuracy, and limitations of SIFT and ORB in real image stitching.

3.METHODOLOGY

The project is written in python programming language. The OpenCV library is used for the image processing. The process starts with choosing the images with identical features and strong overlapping regions. Both SIFT and ORB approach follow the same processing steps: convert images to grayscale, detect key points, compute descriptors, perform descriptor matching using kNN + Lowe's ratio test, compute homography using RANSAC, Warp image 2 into the coordinate frame of image 1 compose the final panorama.

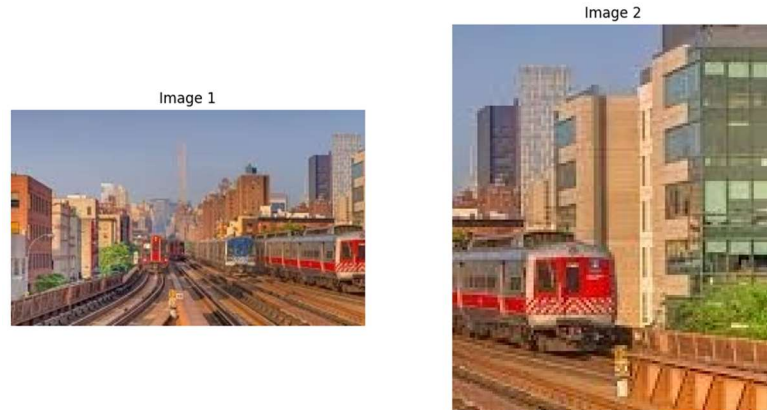


Fig.1 Original images

3.1 SIFT-Based Stitching

The SIFT is implemented based on the classical pipeline introduced by David Lowe and the goal is to extract scale-invariant and rotation-invariant features that provide stable correspondences between the two overlapping images. This begins by detecting the key points using the OpenCV SIFT library and extracting their corresponding 128-dimensional gradient descriptors. FLANN KD-tree matcher is used to perform the descriptor and Lowe's ratio test is used to remove ambiguous matches. The RANSAC method is used to estimate a homography from the remaining matches, and the second image to be warped into the coordinate frame of the first.



Fig.2 SIFT Key points from images

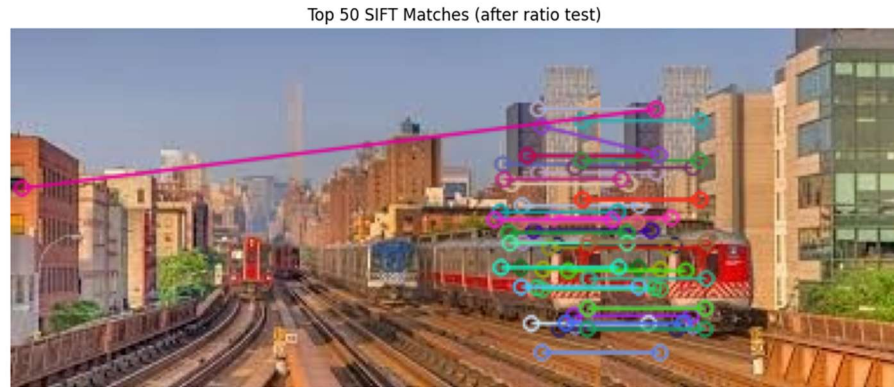


Fig.3 SIFT Key points matching



Fig.4 Final Panorama image from SIFT and Homography

3.2 ORB-Based Stitching

The ORB method is implemented by using OpenCV ORB library and computes 32-byte binary BRIEF descriptors. These descriptors are matched using a brute-force Hamming-distance matcher, with a ratio test applied to refine correspondences. The same RANSAC method is used to compute the homography from the filtered matches, followed by warping and stitching to generate the panorama. Metrics computed for ML Comparison

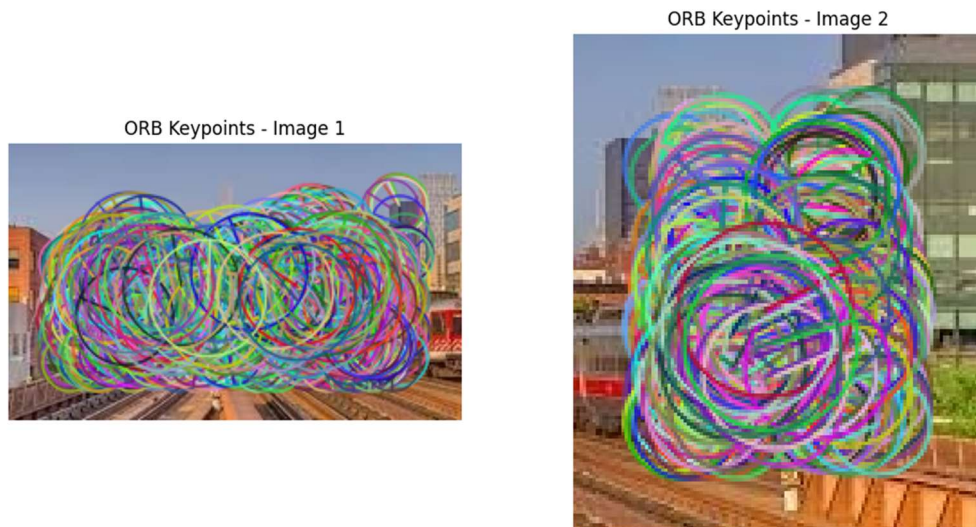


Fig.5 ORB key points from images

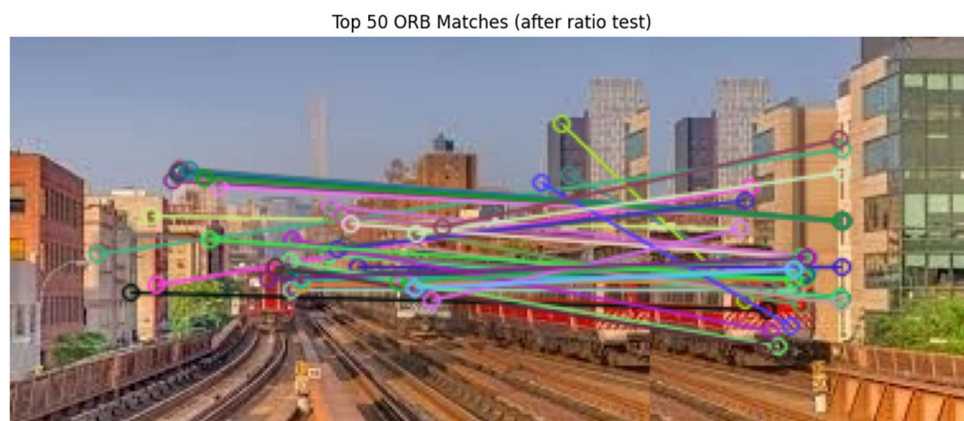


Fig.6 ORB key points matching

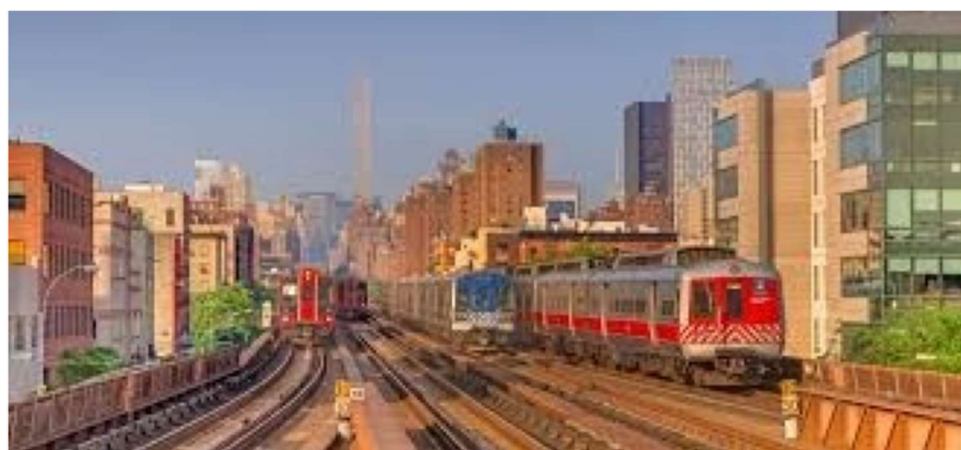


Fig.7 Final Panorama image from ORB and Homography

3.3 Unsupervised ML Evaluation

Several quantitative metrics are extracted from each method to compare the performance of SIFT and ORB methods. Those include the number of good matches, RANSAC inlier ratio, mean descriptor distance, and the mean projection error, which are then combined into feature vectors and normalized before applying K-means clustering. The cluster with high overall performance characteristics was interpreted as the better-performing method. Based on their matching and geometric alignment quality, this unsupervised learning approach provides an objective comparison between SIFT and ORB methods.

4.Results and Discussion

The SIFT implementation produced a high number of key points and reliable feature matches, resulting in a well-aligned and visually consistent panorama, while ORB detected fewer key points overall and generated fewer stable matches leading to a stitched output with small alignment inconsistencies near the overlapping regions. SIFT showed a higher inlier ratio and lower reprojection error, indicating more accurate geometric alignment compared to ORB.

The results showed that the SIFT method produced a larger number of strong matches that survived the ratio test and RANSAC filtering, which contributed to more accurate homography estimation. Even though ORB was fast, it was more sensitive to texture and scale variations. The unsupervised machine learning algorithm comparison validated these results by assigning SIFT to the higher-performing cluster.

5.Conclusion

In conclusion, the ORB remains a faster and lightweight alternative but shows limitations in accuracy under certain conditions, while SIFT provides more stable matches and superior stitching quality for the tested images. The integration of K-means clustering provided an objective, data-driven way to estimate the two methods, confirming SIFT's advantages. This project successfully compared the two classical feature-based methods by using simple machine learning techniques.

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